

25PY201 PHYSICS FOR QUANTUM COMPUTING

Hours Per Week :

L	T	P	SL	C
2	0	2	2	3

PREREQUISITE KNOWLEDGE: Basic mathematical concepts like Linear equations, differential equations and fundamentals of physics and origin of quantum physics.

COURSE DESCRIPTION AND OBJECTIVES:

This course covers the basics of quantum technology and advanced quantum mechanics. It also throws light on fundamentals for quantum computing, and quantum communication.

MODULE-1

UNIT-1

12L+0T+12P+12SL= 36 Hours

INTRODUCTION TO QUANTUM MECHANICS:

Light – matter interaction; Atomic spectra; Bohr's quantization principle; coherence; types of coherence; polarization- types of polarization; Interference, Nonlinear optics. Wave particle duality; postulates of quantum mechanics- quantum states-properties, superposition, entanglement;

UNIT-2

QUANTUM COMPUTATION:

Overview of Quantum Circuits; Deutsch algorithm; Deutsch Josza algorithm; Database search- Grover's algorithm; Shor's Algorithm. Quantum Circuit Model; Quantum computing Hardware (Superconducting Qubits, Ion Traps);

PRACTICES:

- Visualize qubit state vectors and rotations (Bloch Sphere Visualizer).
- Simulate Bell states and measure correlation (Entangled States and Correlations).
- Analysing quantum circuits and gate operations.
- Simulating quantum systems using software (Qiskit, Cirq).
- Deriving quantum algorithms (e.g. Shor's, Grover's).

MODULE-2

UNIT-1

18L+0T+18P+27= 63 Hours

QUANTUM INFORMATION THEORY:

Classical bits vs Quantum bits (Qubits); Bell inequalities; Di Vincenzo criteria for realizing qubits; Quantum Computing vs Classical Computing; Q-gates, Universal quantum logic gates; quantum teleportation; Superdense coding; Quantum Cryptography (BB84 Protocol); Challenges of quantum computers; Quantum Information Processing; quantum error correction codes.

UNIT-2

QUANTUM COMPUTATION:

Overview of Quantum Circuits; Deutsch algorithm; Deutsch Josza algorithm; Database search- Grover's algorithm; Shor's Algorithm. Quantum Circuit Model; Quantum computing Hardware (Superconducting Qubits, Ion Traps); .

UNIT-3**INTEGRATIONS:****PRACTICES:**

- Visualize qubit state vectors and rotations (Bloch Sphere Visualizer).
- Simulate Bell states and measure correlation (Entangled States and Correlations).
- Analysing quantum circuits and gate operations.
- Simulating quantum systems using software (Qiskit, Cirq).
- Deriving quantum algorithms (e.g. Shor's, Grover's).

COURSE OUTCOMES:

Upon successful completion of this course, students will be able to:

CO No.	Course Outcomes	Bloom's Level	Mapping with POs
1	Apply the fundamental principles of quantum mechanics, including light-matter interaction, polarization, and coherence. Apply quantum postulates and mathematical tools like operators, commutators, and eigenvalue equations.	Apply	1(3), 2(3)
2	Analyze experimental quantum phenomena (e.g., photoelectric effect, Young's experiment) and simulate quantum states.	Analyze	1(3), 2(3)
3	Distinguish between classical and quantum information concepts and evaluate the properties of qubits and gates.	Evaluate	1(3), 2(3), 4(3)
4	Analyze quantum circuits using algorithms like Deutsch, Grover, and Shor with real-world quantum hardware awareness.	Analyze	1(3), 2(3), 5(2),

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):**TEXT BOOKS:**

1. Introduction to Quantum Mechanics, David J. Griffiths, Cambridge University Press, 3rd edition, 1018.
2. Quantum computation and quantum information by Nielsen and Chuang.
3. Chris Bernhardt, Quantum Computing for Everyone, The MIT Press, Cambridge, 2020.
4. Quantum information theory by Mark M. Wilde.

REFERENCE BOOKS:

1. Chuck Easttom, "Quantum Computing Fundamentals", Addison-Wisley, 2021.
2. Giuliano Benenti, Giulio Casati and Giuliano Strini, "Principles of Quantum Computation and Information", Volume I: Basic Concepts, "World Scientific", 2004
3. Leonard Susskind and Art Friedman, Quantum Mechanics: The Theoretical Minimum, Basic Books, Perseus Books Group 2015. ISBN 13: 978 0 465 06290 4
4. Eleanor G. Rieffel & Wolfgang H. Polak, Quantum Computing: A Gentle Introduction, The MIT Press (Scientific and Engineering Computation series), 2011. ISBN 13 978 0262015066
5. Modern Quantum Mechanics, J. J. Sakurai and Jim Napolitano, Cambridge University Press, 2nd edition, 2017.
6. Quantum Computing, A Gentle Introduction, Eleanor G. Rieffel and Wolfgang H. Polak MIT

SKILLS:

- ✓ Apply the dynamics of Light to realize the various potential applications in Engineering.
- ✓ Evaluate the concepts of Lasers and Optical Fibers to realize versatile applications in Science, Engineering and Technology.
- ✓ Analyze the Crystal Structures and orientation of planes.
- ✓ Appraise the importance of Ultrasonics in medicine.
- ✓ Demonstrate the synthesis and characterization of Nano materials in view of their applications.